# 1 FROM SCARCE TO AN *URAVU:* EXTRACTING POTABLE WATER FROM THE AIR!

As the world sees drinking water sources become scarce amid unprecedented climate changes, Bengaluru-based startup Uravu Labs is creating water out of thin air through renewable energy sources


*Uravu water factory*

Star Wars legend Luke Skywalker's hot desert planet, Tatooine, had a moisture farm that used "vaporators" to pull drinking water from the air. For a long time, vaporators had remained a figment of science fiction until two NIT Calicut graduates faced water rationing in college after the waterbody which supplied water to their campus dried up.

Propelled by a looming water crisis, Swapnil Shrivastav and Venkatesh R decided to turn fiction into reality in 2016. They didn't realise that six years later, they would have managed to achieve it with Uravu Labs. *Uravu* means a spring of water in Malayalam.

In 2017-18, the duo was joined by Govinda Balaji and Pardeep Garg. Together, the futuristic four designed a device to capture water from the air using a hygroscopic substance called a desiccant with a saltwater solution called brine. The air is passed over the brine, which absorbs the moisture, which, in turn, saturates the solution. The brine is then heated with solar energy to evaporate the water, and the resulting water vapour is collected.

The machine is equipped with various sensors to monitor five main parameters. "We monitor the relative humidity of the ambient air inside the device before condensation and the temperature of the air and of the water being formed. Flow rates are also measured at different points, including air and water flow rates. To keep track of the overall performance, we also have a sensor to measure the amount of water produced by the system. Lastly, we ensure water quality by monitoring pH and total dissolved solids (TDS) levels in the water output," elaborates Shrivastav.

Initially, the data from these sensors was gathered using a Raspberry Pi and visualised through an AWS-based dashboard for analysis. As the startup progressed towards an industrial-grade setup in 2021-22, they transitioned to using programmable logic controllers (PLCs) for control and data acquisition.

"The innovation in our electronics side is centred around real-time problem solving. We need to know the exact conditions, such as humidity levels, water production, etc., in real time. This demands a lot of decision-making and analysis on the board level, and the robust electronics hardware gives us more control and real-time responsiveness in our water capture devices," explains Shrivastav.